

Ray Tsai

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EDUCATION

University of Oxford

Oxford, United Kingdom

M.Sc. in Mathematical Sciences

Sep 2025 – Jun 2026

- **Dissertation:** *Induced Subgraphs in Ramsey Graphs*, advised by Prof. Alex Scott
- **Relevant coursework:** Theories of Deep Learning, Probability on Graphs and Lattices, Probabilistic Combinatorics, Networks, Continuous Optimization

University of California San Diego

La Jolla, CA

B.Sc. in Mathematics-Computer Science (Honors with Highest Distinction), Math GPA: 3.98/4.0 Sep 2021 – Jun 2025

- **Honors Thesis:** *Double Turán Problems*, advised by Prof. Jacques Verstraete
- **Graduate coursework:** Statistical Learning, Combinatorics, Discrete Geometry, Complex Analysis

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript/TypeScript, L^AT_EX

Libraries & Tools: PyTorch, NumPy, NextJS, Astro, Git, React/React Native, Jupyter

PROJECTS

Adversarial Robustness via Smooth Activations | *PyTorch, Python*

Oxford, 2026

- Proposed a unified theoretical view of FGSM and Input Gradient Regularization, explaining why smooth activations (GeLU, Softplus) prevent catastrophic overfitting in one-step adversarial training.
- Empirically confirmed that smooth activations recover near-full robustness at a fraction of the cost of standard adversarial training, extending prior results to new activation functions.

WORK EXPERIENCE

MSI

New Taipei City, Taiwan

Software Engineer Intern

Jul 2023 – Aug 2023

- Fine-tuned Stable Diffusion model in PyTorch to generate realistic MSI product images for marketing; integrated model into *MSI AI Artist*, a Windows desktop application for local AI image generation.
- Engineered automated installation pipeline for AI inference optimization backends, improving image generation throughput by 80%.

Kook

San Diego, CA

Co-founder & Lead Software Engineer

Dec 2021 – Dec 2022

- Co-founded hardware-software startup for automated surfboard rental stations; secured UCSD Basement incubation and led 5-member engineering team.
- Built full-stack mobile application in React Native with SQL backend and NFC-based authentication, reducing rental initiation time by 75%; developed embedded firmware in C++ on Arduino for hardware lock control.

RESEARCH EXPERIENCE

UC San Diego, Department of Mathematics

La Jolla, CA

Undergraduate Research Assistant

Mar 2023 – Jun 2025

- Conducted independent research in extremal graph theory under Prof. Jacques Verstraete; results on *Double Turán Problems* constitute a manuscript in preparation for publication.
- Earned highest distinction in the UCSD Math Honors Program (awarded to ~12 students annually).

UC San Diego, Department of CSE

La Jolla, CA

Undergraduate Research Assistant

Apr 2024 – Jun 2025

- Studied graduate-level complexity theory and joined Prof. Russell Impagliazzo's research group on smooth boosting; explored connections between algorithmic fairness, computational complexity, and learning theory.

AWARDS

Physical Sciences Dean's Undergraduate Award for Excellence

UC San Diego, 2025

- Awarded to top undergraduates in Mathematics, Chemistry & Biochemistry, and Physics for academic excellence and research promise. (\$1000)